

MockStar: AI Copilot Platform for Interview Preparation

Adithya Subramaniam Harshit Kaushik Vijay Kumar Prakash

Project Overview

Our capstone project is to develop a web-based AI interview preparation platform that simulates realistic interview interactions and provides personalized feedback based on both verbal and non-verbal communication. Our system integrates LLMs for dynamic interview question generation, speech analysis for communication feedback, and computer vision techniques to analyze visual cues such as gaze and facial engagement. The platform is designed with accessibility in mind and aims to support a diverse range of users, including individuals who may lack access to structured interview coaching.

Progress So Far:

Problem Definition and Research

We conducted initial research (six user interviews) into the limitations of existing interview preparation tools, such as static question banks and text-based coaching platforms. We identified key gaps in the current landscape, including the lack of multimodal feedback, limited personalization, costly onboarding if one were to train their own model, and insufficient accessibility considerations. This research informed the overall project direction and the decision to build a multimodal interview simulation platform.

MVP and Scope Definition

We also defined the core features of the minimum viable product (MVP), including:

- An interactive AI interviewer capable of generating interview questions dynamically
- A web interface that allows users to participate in mock interviews using video and audio
- Speech analysis capabilities to detect filler words, pacing, and communication clarity
- Computer vision features to evaluate non-verbal cues such as gaze and facial engagement
- A feedback system that provides users with actionable suggestions for improving their interview performance

System Architecture Planning

We have begun defining the platform's overall system architecture. The planned architecture includes:

- A web-based frontend (React) that manages user interaction, video/audio capture, and display of interview prompts and feedback.

- A Python-based backend implemented using FastAPI to handle session management, API communication, and orchestration of different processing components. FastAPI was chosen for its lightweight design and strong support for asynchronous workflows.
- Integration with large language models (OpenAI APIs) to generate interview questions and produce personalized feedback summaries.
- A speech processing pipeline using Whisper for transcription, followed by simple analytics to detect filler words, speaking rate, and pauses.
- Computer vision modules using OpenCV and MediaPipe for face detection and basic gaze estimation to approximate user engagement during responses.

Pending architectural questions include: (i) should we have separate models for audio and video, or should they all run using the same LLM under the hood, (ii) should we run the application via a browser or do we opt for local application installation and execution using offline LLMs (ollama) and (iii) do we need to set up an orchestrator agent & worker agents setup. We plan to consult with professionals before making a final decision here.

Role Allocation and Team Responsibilities

We have divided responsibilities amongst ourselves to ensure equal distribution of work across technical and product components:

Adithya - Speech processing and multimodal language integration

Harshit - Computer vision implementation for visual analysis

Vijay - Product design, system integration, and project coordination

Initial Technical Exploration

We have begun exploring relevant tools and frameworks for key system components. This includes evaluating speech transcription models, computer vision libraries for facial and gaze detection, and potential backend frameworks for integrating the multimodal pipelines. These explorations are informing the technical stack and helping us assess feasibility for the planned features.

We have working code for the nascent stages of individual components (e.g., audio-only) that we will demo with our advisor at our next meeting.

Work Remaining Before Final Deliverable

The remaining work focuses on building, integrating, and evaluating the core system components required for the final prototype.

- **Frontend Development:** a web interface must be implemented that allows users to participate in a simulated interview session. This includes enabling webcam and

microphone input, displaying interview prompts, and presenting feedback after the session.

- **Backend System Implementation:** the backend will handle session management, communication between system components, and coordination of multimodal data processing. This includes building APIs that connect the frontend interface with the speech processing, computer vision, and language model components.
- **Speech Processing Pipeline:** the system will incorporate speech recognition and analysis features that extract meaningful communication metrics from user responses. These include transcription, filler word detection, speaking rate analysis, and pause detection.
- **Computer Vision Module:** the vision component will analyze the user's video stream to estimate visual engagement. Planned features include face detection, head pose estimation, and approximate gaze direction to identify patterns such as looking away frequently during responses.
- **Multimodal Feedback Generation:** data collected from the speech and vision modules will be aggregated and passed to a feedback generation component powered by a language model. This component will generate structured coaching feedback summarizing strengths, areas for improvement, and recommended practice strategies.
- **Personalization and Session Tracking:** the system will maintain session history and performance metrics to provide users with longitudinal insights into their progress across multiple practice interviews.
- **Accessibility Features:** the platform will incorporate accessibility considerations aligned with WCAG 2.2 AA guidelines, including readable interface design, transcript support, and flexible interaction modes.
- **User Testing and Evaluation:** we plan to conduct usability testing with a small group of participants to evaluate the effectiveness of the platform and get early feedback on the design. The evaluation will focus on user experience, perceived usefulness, and potential improvements.
- **Final Documentation and Presentation:** the final phase will involve preparing the project report, documenting system architecture and implementation details, summarizing evaluation findings, and developing a demonstration of the working prototype.

Project Trajectory

We believe we are on track to deliver a working prototype by the end of April or early May. With foundational design work and early technical exploration completed so far provide a clear roadmap for the remaining implementation and evaluation phases.